A Linear-Fit Micro-Experiment as a Research OS Smoke Test

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Abstract

This note is the worked example for the AI-Human Research OS. It walks one idea through the full loop — OKF idea concept, project instantiation from the template, a runnable experiment, a generated figure, and a compiled PDF — with every number traceable to code in `Code/`.

1 Setup

We generate $N = 50$ points from $y = 2x + 1 + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.5^2)$ and a fixed seed (42), then fit a degree-1 polynomial by least squares (`numpy.polyfit`). The script is `Code/fit_line.py`; run it with `uv run python fit_line.py`. The citation pipeline is exercised with a real reference fetched from the arXiv API [1].

2 Result

The fit recovers the generating parameters: fitted slope 2.0452 (true 2.0), fitted intercept 0.9325 (true 1.0), MSE = 0.1402 (script output, 2026-06-12). Figure 1 shows the data and the fitted line.

References

[1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2017.

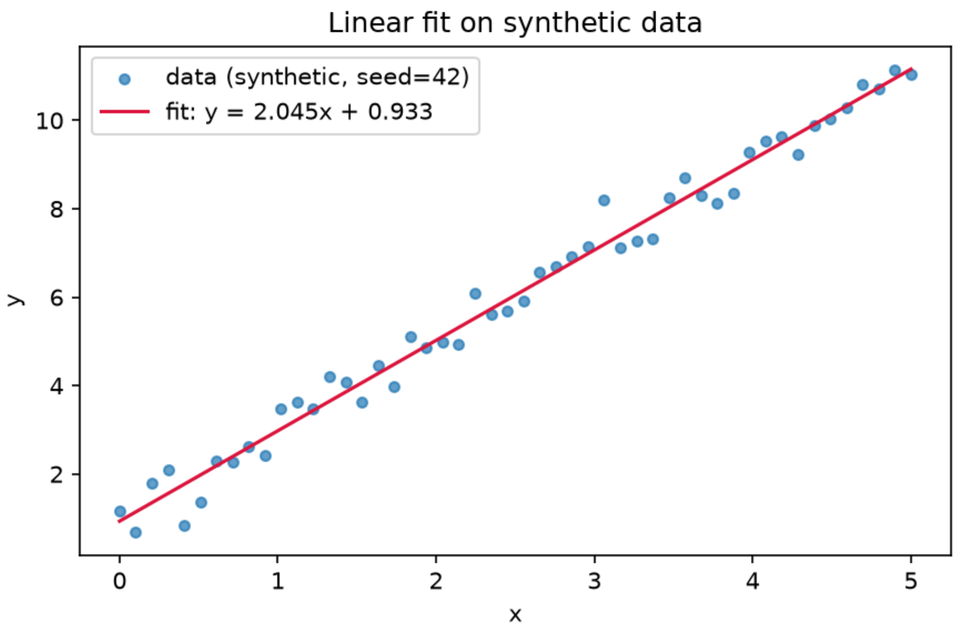

Figure 1: Synthetic data and least-squares linear fit (seed 42).